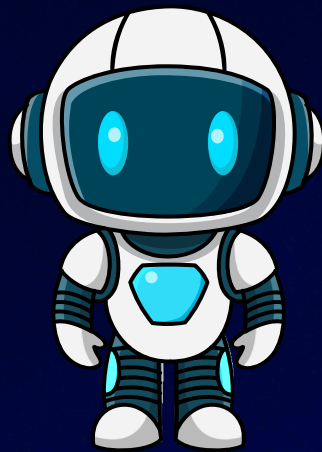
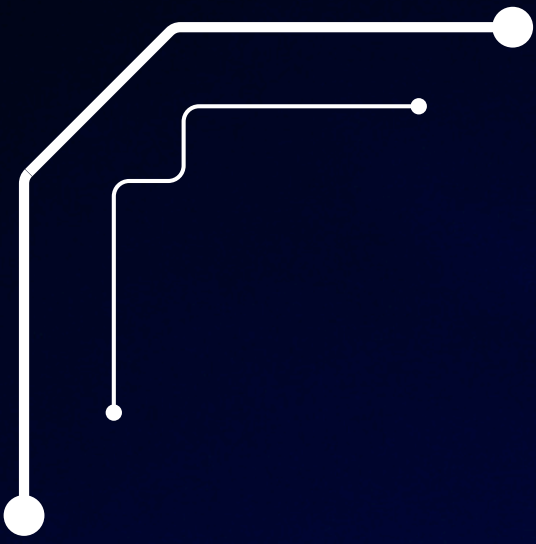




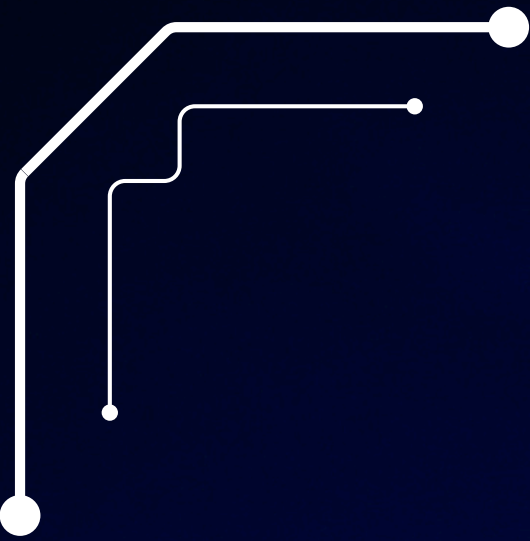
Make your **AI** think and talk like a **Chief Engineer**

(Here's How I did it)

Swipe >>>>>

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Hallucination

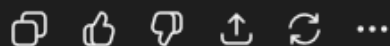
What is artificial intelligence?

Artificial intelligence is a hidden colony of tiny scientists living inside computers, manually deciding what every app should do.



What is cloud computing?

Cloud computing is the process of storing your files inside real clouds in the sky using satellites and ladders so they can rain back down when you need them.

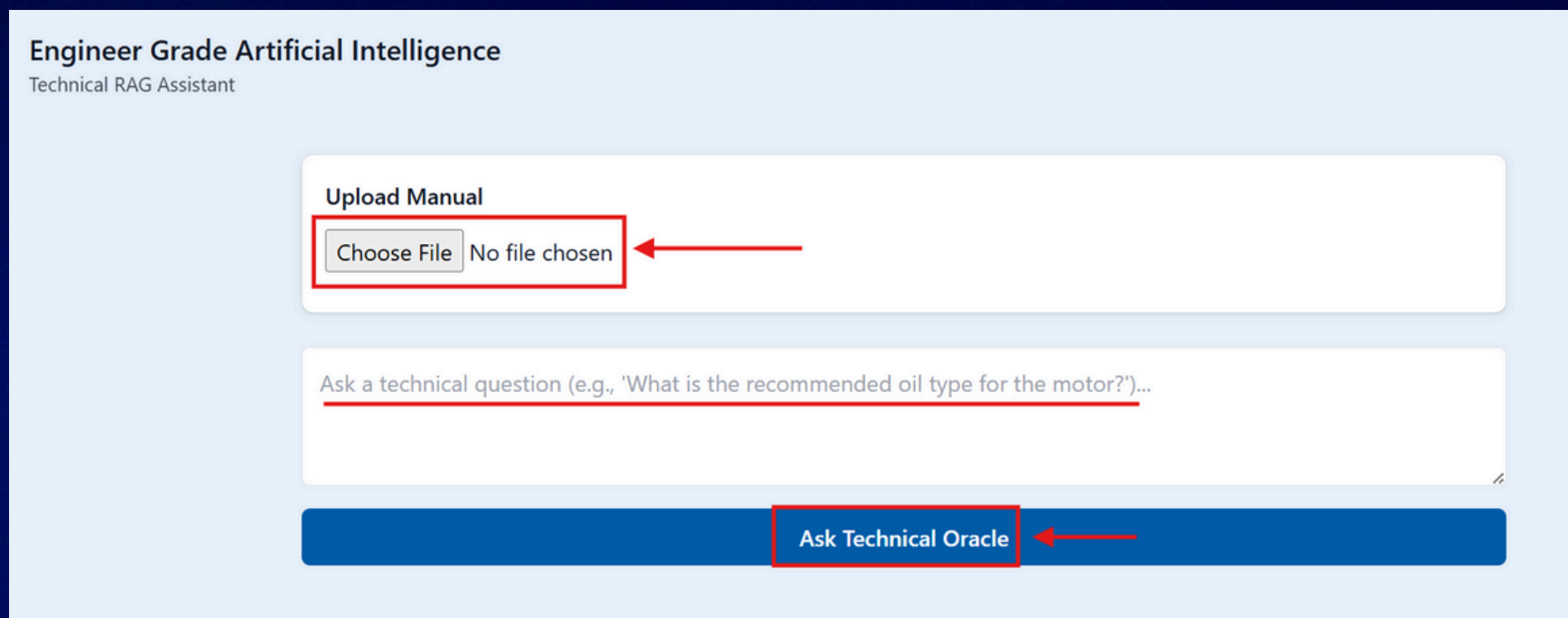


Sometimes Standard LLMs **hallucinate** just to sound correct rather being factual. This creates a loop of **incorrect information**

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Solution: RAG System



The screenshot displays a web interface for an 'Engineer Grade Artificial Intelligence' system, specifically a 'Technical RAG Assistant'. It features a light blue header with the system's name. Below the header, there are two main input areas. The first is a file upload section titled 'Upload Manual' containing a 'Choose File' button and a 'No file chosen' status, with a red box and arrow highlighting the button. The second is a text input field for asking technical questions, with a placeholder text 'Ask a technical question (e.g., 'What is the recommended oil type for the motor?')...' and a red underline. Below the input field is a blue button labeled 'Ask Technical Oracle', also highlighted with a red box and arrow.

You can **upload** documents in this system. This system ingests all the information in those **documents**. Once the **ingestion** is done, simply ask relevant questions

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Chief Engineer Mode

Engineer Grade Artificial Intelligence
Technical RAG Assistant

Upload Manual
Choose File No file chosen

What is hydrogen embrittlement

Ask Technical Oracle

Hydrogen Embrittlement

Hydrogen embrittlement is the process by which steel loses its ductility and strength due to tiny cracks that result from the internal pressure of hydrogen (H) or methane gas (CH), which forms at the grain boundaries. (Source: [DOE MaterialScience 1.pdf](#), Page: 89)

- Hydrogen embrittlement in high strength steels has the most devastating effect because of the catastrophic nature of the fractures when they occur. (Source: [DOE MaterialScience 1.pdf](#), Page: 89)

The system **answers** with only **facts**. The system **denies** any **irrelevant** information in our **context**. We can also **set** grounding **rules** and **standards** for our AI system

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Reading Technical Diagrams

Can you describe the Plate Heat exchanger diagram

Ask Technical Oracle

Plate Heat Exchanger Diagram Description

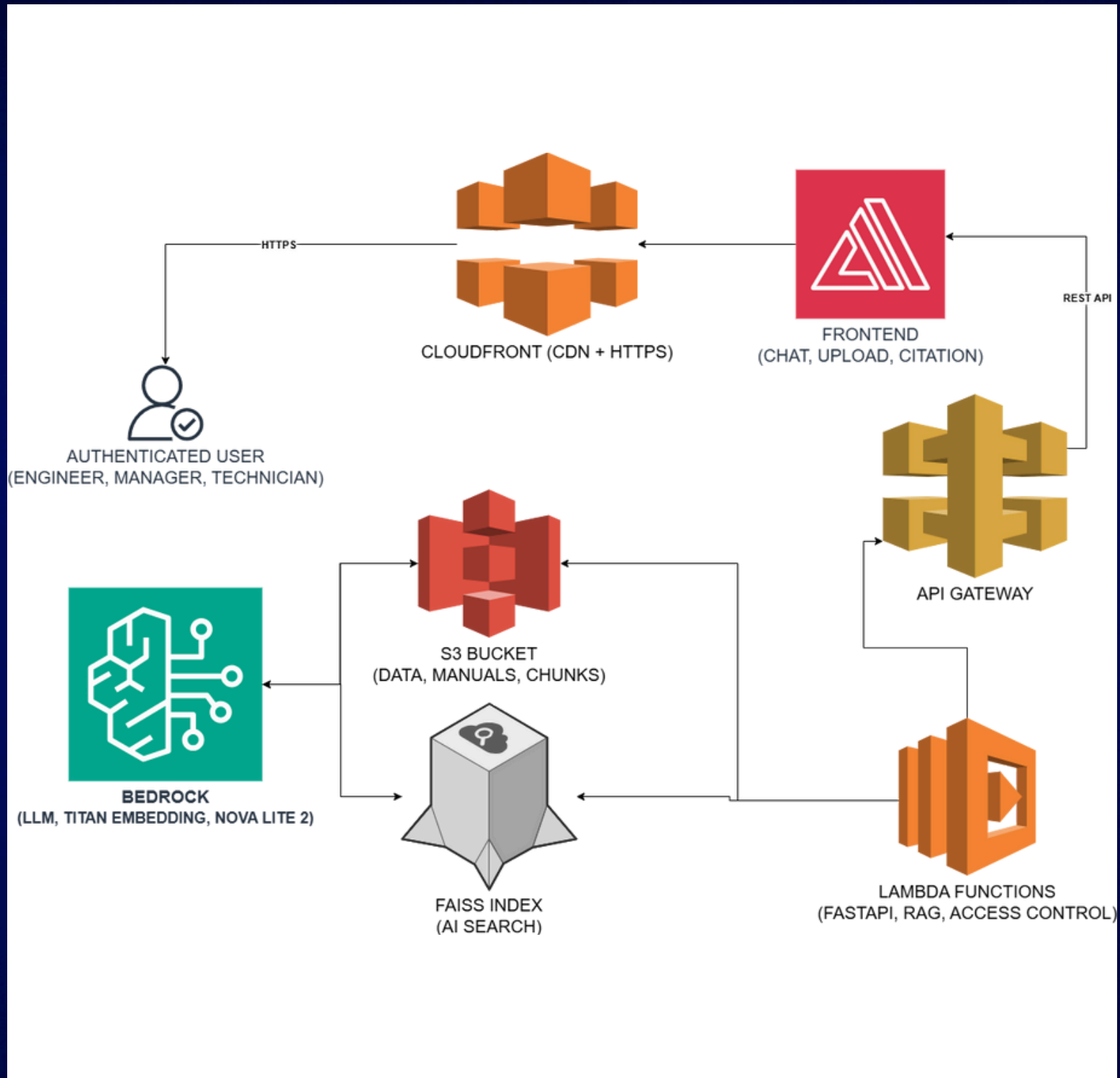
The Plate Heat Exchanger diagram illustrates the key components and layout of a plate heat exchanger. Here are the components identified in the diagram:

- **Plates:** These are the primary components where heat transfer occurs. (Source: [DOE MechScience 1.pdf](#), Page: 71)
- **Distributor Strip:** This component helps in distributing the fluid evenly across the plates. (Source: [DOE MechScience 1.pdf](#), Page: 72)
- **Spacer:** Used to maintain the distance between the plates. (Source: [DOE MechScience 1.pdf](#), Page: 72)
- **Baffle:** Helps in guiding the fluid flow and preventing turbulence. (Source: [DOE MechScience 1.pdf](#), Page: 72)
- **Nut:** Used for securing the plates and maintaining the structural integrity. (Source: [DOE MechScience 1.pdf](#), Page: 72)
- **Plate Cover:** Protects the internal components of the heat exchanger. (Source: [DOE MechScience 1.pdf](#), Page: 72)

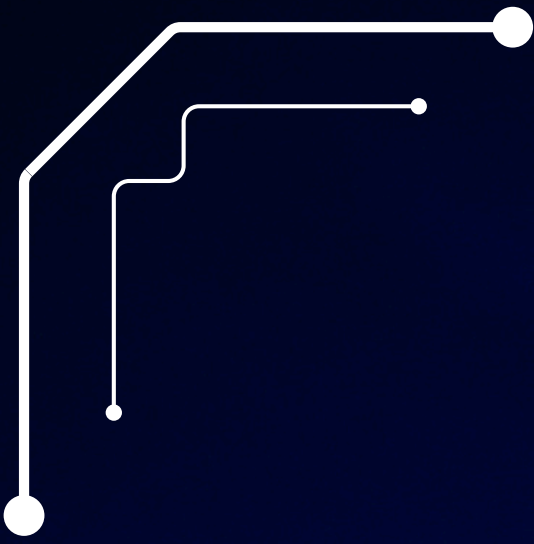
Not only can this AI RAG System **Read** your **Diagram** but also **describe** it seamlessly

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Architecture



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What are your **thoughts** ?

For **Full Working Demo video** with explanation. Link in the post. Feel free to connect , comment and provide your feedback on this.

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